

Multimodal Crop Disease Classification Using Spectral Data

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Abstract

Crop disease detection is an important task in precision agriculture, as early identification of diseases can significantly improve crop yield and reduce economic losses. In this project, we address the problem of classifying wheat crops into three categories: Health, Rust, and Other using multimodal UAV imagery. The dataset consists of RGB, multispectral (MS), and hyperspectral (HS) data, where each modality captures different levels of spectral information.

We propose a multimodal framework that combines multispectral and hyperspectral features using a gated fusion mechanism and cross-attention. The fused features are used for classification using both a neural network classifier and classical machine learning models such as Support Vector Machine (SVM), Random Forest, Logistic Regression, and K-Nearest Neighbors (KNN). Experimental results show that multispectral data achieves the highest accuracy, while the fusion model provides competitive performance. The study demonstrates that spectral data plays a crucial role in improving crop disease classification.

1 Introduction

Crop disease detection is a fundamental problem in modern agriculture, as plant diseases can significantly affect productivity and food security. Early and accurate identification of diseases enables timely intervention, reducing crop loss and improving yield. Traditionally, disease detection has relied on manual inspection or analysis of RGB images. However, RGB images are limited to the visible spectrum and often fail to capture subtle physiological changes in plants, especially in early stages of infection.

Recent advancements in remote sensing and imaging technologies have introduced multispectral and hyperspectral data for agricultural applications. Multispectral data captures a small number of carefully selected spectral bands, including red-edge and near-infrared (NIR), which are highly sensitive to plant health and chlorophyll content. Hyperspectral data, on the other hand, provides a dense representation of the spectral signature across a wide range of wavelengths. These modalities enable the detection of plant stress and disease characteristics that are not visible in standard RGB images.

In this project, we aim to leverage the complementary strengths of multispectral and hyperspectral data through a multimodal learning framework. We design modality-specific encoders to extract features from multispectral and hyperspectral inputs, followed by a gated fusion mechanism and cross-attention to combine the features effectively. The resulting fused representation is used for classification into three categories: Health, Rust, and Other.

Furthermore, we explore two classification strategies: (1) an end-to-end neural network classifier applied directly on the fused features, and (2) classical machine learning classifiers such as SVM, Random Forest, Logistic Regression, and KNN applied on extracted embeddings. We also compare the performance of individual modalities (RGB, MS, HS) with the fusion approach to evaluate the effectiveness of multimodal learning.

The results demonstrate that multispectral data provides strong discriminative features for disease classification, while the fusion model achieves competitive performance by combining complementary information from multiple modalities. This highlights the importance of spectral data in agricultural disease detection tasks.

2 Literature Review

Crop disease detection has been widely studied using both traditional machine learning and deep learning approaches. Early methods primarily relied on handcrafted features extracted from RGB images, followed by classifiers such as Support Vector Machines (SVM) and Random Forests. While these methods achieved moderate success, their performance

was limited due to the lack of rich feature representation and sensitivity to lighting and background variations.

With the advancement of deep learning, Convolutional Neural Networks (CNNs) have become the dominant approach for image-based plant disease classification. CNN-based models trained on RGB images have demonstrated improved performance by automatically learning spatial features such as texture and color patterns. However, RGB images are restricted to the visible spectrum and often fail to capture subtle physiological changes in plants, especially in early stages of disease.

To overcome these limitations, researchers have explored the use of multispectral (MS) and hyperspectral (HS) data. Multispectral imaging captures a small number of carefully selected spectral bands, including red-edge and near-infrared (NIR), which are highly sensitive to plant health. Hyperspectral imaging provides a much denser spectral representation across a wide range of wavelengths, allowing for detailed analysis of plant physiology. Studies have shown that models using spectral data generally outperform RGB-based models, with reported accuracies often ranging between 75% and 90%.

Despite these advancements, many existing works focus on a single modality, either RGB, multispectral, or hyperspectral data. Limited work has been done on effectively combining multiple modalities to leverage their complementary strengths. In addition, most studies rely only on deep learning classifiers, without exploring the potential of classical machine learning methods on learned feature representations.

In contrast to existing approaches, this project proposes a multimodal framework that combines multispectral and hyperspectral data using feature-level fusion. Furthermore, both neural network classifiers and classical machine learning models are evaluated on the fused features. This provides a more comprehensive understanding of how different modalities and classification strategies contribute to performance.

3 Proposed Methodology

In this section, we describe the proposed approach for crop disease classification using multimodal data. The overall framework consists of data preprocessing, feature extraction from different modalities, feature fusion, and classification.

3.1 Dataset and Data Representation

The dataset consists of three aligned modalities: RGB, multispectral (MS), and hyperspectral (HS) images. Each sample corresponds to the same physical location, ensuring spatial

alignment across modalities. The classification task involves three categories: Health, Rust, and Other.

RGB images contain three channels corresponding to the visible spectrum. Multispectral data consists of five important spectral bands: blue, green, red, red-edge, and near-infrared (NIR). Hyperspectral data contains a large number of bands (approximately 125), covering a wide spectral range from 450 nm to 950 nm. While hyperspectral data provides rich information, it also introduces noise and high dimensionality.

3.2 Feature Extraction

Separate encoders are used to extract features from multispectral and hyperspectral inputs. These encoders learn meaningful representations from each modality independently.

The multispectral encoder focuses on key vegetation-sensitive bands such as red-edge and NIR, which are highly informative for detecting plant stress and disease. The hyperspectral encoder processes a large number of spectral bands to capture detailed spectral signatures of plant health.

The output of each encoder is a feature vector representing the corresponding modality.

3.3 Feature Fusion

To combine information from different modalities, a feature-level fusion strategy is used. First, the features from multispectral and hyperspectral encoders are weighted using a gating mechanism. The gating mechanism assigns learnable importance weights to each modality.

After gating, cross-attention is applied to model interactions between modalities. This allows the model to learn relationships between multispectral and hyperspectral features rather than treating them independently.

Finally, the fused features are aggregated into a compact 256-dimensional embedding, which serves as a unified representation of the input data.

3.4 Classification Strategies

Two classification approaches are explored in this project:

3.4.1 Neural Network Classifier

The fused feature vector is passed through a fully connected neural network layer to perform classification. The model is trained in an end-to-end manner using a standard loss function such as cross-entropy loss.

3.4.2 Classical Machine Learning Classifiers

In addition to the neural network classifier, classical machine learning models are applied on the extracted fusion embeddings. These include:

- Support Vector Machine (SVM)
- Logistic Regression
- Random Forest
- K-Nearest Neighbors (KNN)

This approach allows us to evaluate how well the learned features perform when used with different types of classifiers.

3.5 Baseline Models

To evaluate the effectiveness of the proposed fusion approach, single-modality models are also implemented:

- RGB-based model
- Multispectral model
- Hyperspectral model

These baseline models provide a reference for comparing the performance of the multi-modal fusion framework.

4 Experimental Results

In this section, we evaluate the performance of the proposed multimodal framework and compare it with single-modality models. The evaluation is carried out using standard metrics and visual analysis.

4.1 Dataset Description

The dataset used in this study consists of UAV-acquired wheat images with three aligned modalities: RGB, multispectral (MS), and hyperspectral (HS). Each sample belongs to one of three classes: Health, Rust, or Other.

The multispectral data contains five important spectral bands, including red-edge and near-infrared (NIR), which are highly sensitive to plant health. The hyperspectral data contains a large number of bands (approximately 125), covering a wide spectral range. While hyperspectral data provides rich information, it also introduces noise and high dimensionality, which can affect model performance.

The dataset is divided into training and validation sets. The training set is used to train the models, while the validation set is used to evaluate performance.

4.2 Experimental Settings

The models are trained using standard deep learning techniques. A neural network classifier is applied on the fused features, and classical machine learning models such as SVM, Logistic Regression, Random Forest, and KNN are applied on the extracted embeddings.

The evaluation is performed using accuracy and F1-score metrics. Accuracy measures the overall correctness of the model, while F1-score provides a balanced evaluation of precision and recall across different classes.

4.3 Performance Comparison

To evaluate the effectiveness of different modalities, we compare the performance of RGB, multispectral, hyperspectral, and fusion-based models.

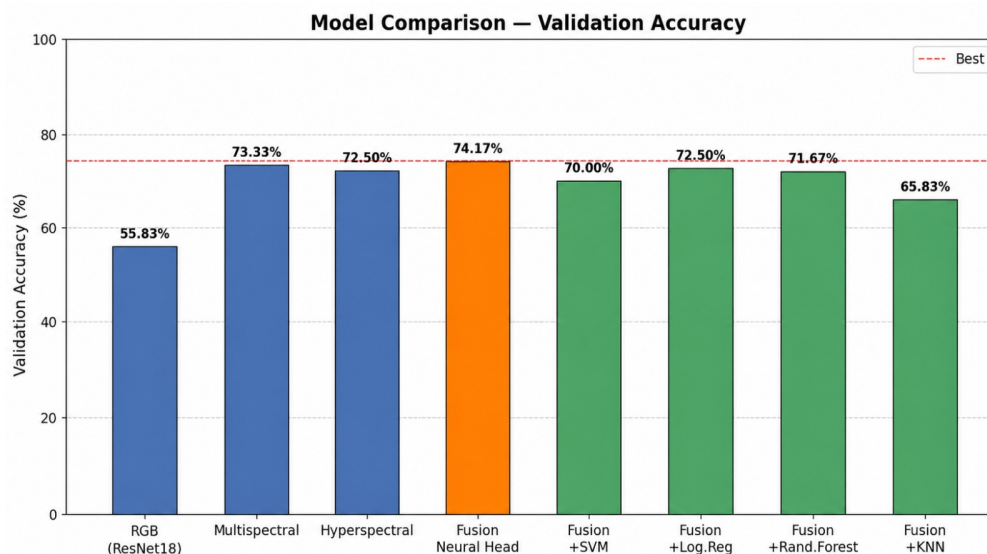


Figure 1: Validation accuracy comparison across different models.

The fusion neural head achieves the highest validation accuracy (74.17%), slightly out-

performing the multispectral model (73.33%), while the hyperspectral model also performs competitively (72.50%). This can be attributed to the fact that multispectral data contains key spectral bands, such as red-edge and near-infrared, which are highly informative for detecting plant stress and disease.

The RGB model achieves the lowest accuracy, as it is limited to the visible spectrum and cannot capture subtle physiological changes in plants. The hyperspectral model performs better than RGB but slightly lower than multispectral due to the presence of noise and high dimensionality.

The fusion model achieves competitive performance by combining information from multispectral and hyperspectral data. However, the inclusion of hyperspectral data does not significantly improve performance over the multispectral model, indicating that hyperspectral features require more careful preprocessing.

4.4 Confusion Matrix Analysis

To further analyze the performance of the models, confusion matrices are used to visualize classification results for each class.

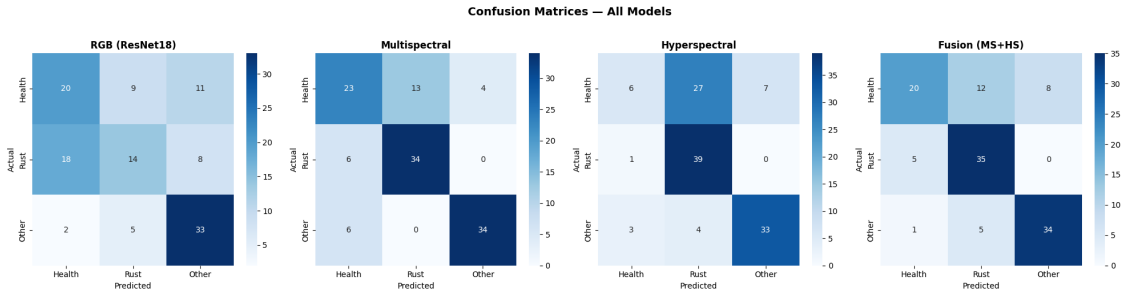


Figure 2: Confusion matrices for RGB, multispectral, hyperspectral, and fusion models.

From the confusion matrices, it can be observed that the multispectral model shows strong diagonal values, indicating accurate classification across all classes. In contrast, the RGB model shows higher misclassification, particularly between Rust and Other classes.

The fusion model improves classification compared to RGB and hyperspectral models, but its performance remains close to the multispectral model.

4.5 F1-Score Analysis

F1-score is used to evaluate the performance of the model across individual classes.

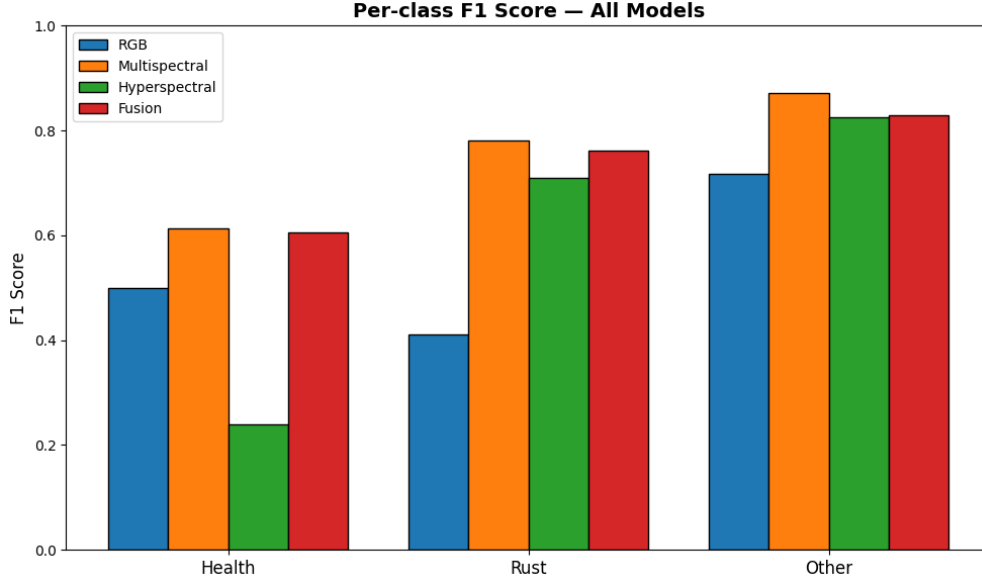


Figure 3: Per-class F1-score comparison across different models.

The results indicate that the multispectral model achieves balanced performance across all classes. The fusion model also shows improved class-wise balance compared to RGB and hyperspectral models.

4.6 Fusion Model Analysis

To understand how the fusion model combines information from different modalities, we analyze the learned gate weights.

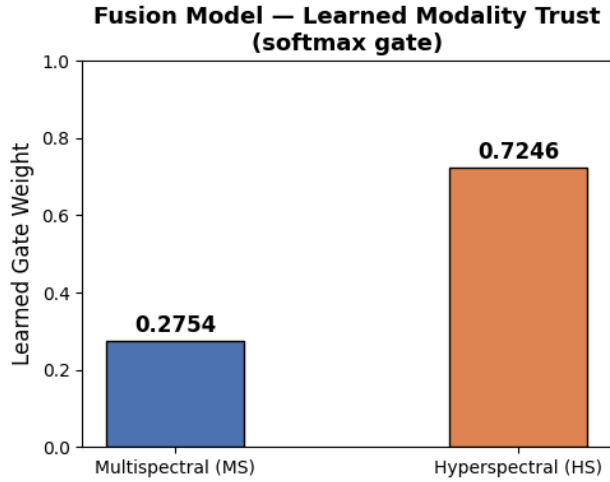


Figure 4: Learned gate weights for multispectral and hyperspectral modalities.

The gate weights indicate that the model assigns higher importance to hyperspectral

features. However, despite this higher weight, the overall performance suggests that multispectral data provides more stable and discriminative features.

This observation highlights that while hyperspectral data is rich in information, it may also contain noise, and effective preprocessing is necessary to fully utilize its potential.

4.7 Comparison with Existing Methods

Previous studies using RGB-based models report moderate accuracy due to limited spectral information. In contrast, methods using multispectral and hyperspectral data achieve higher accuracy, typically in the range of 72% to 81%.

The results of this project are consistent with these findings, as multispectral and fusion-based models outperform the RGB model. This demonstrates the importance of spectral data in improving crop disease classification.

4.8 Time Complexity

The fusion model introduces additional computational cost due to the use of multiple encoders and attention mechanisms. In contrast, classical machine learning models applied on extracted embeddings are computationally efficient.

Overall, the proposed approach balances performance and computational complexity by combining deep learning and classical machine learning techniques.

5 Summary

In this project, we investigated crop disease classification using multimodal UAV imagery consisting of RGB, multispectral (MS), and hyperspectral (HS) data. The objective was to classify wheat samples into three categories: Health, Rust, and Other.

We designed a multimodal framework that extracts features from multispectral and hyperspectral inputs using separate encoders and combines them using a gated fusion mechanism with cross-attention. The fused representation is used for classification through two approaches: an end-to-end neural network classifier and classical machine learning models applied on extracted embeddings.

Experimental results show that the multispectral model achieves the highest accuracy due to the presence of key vegetation-sensitive bands such as red-edge and near-infrared. The hyperspectral model, while rich in information, is affected by high dimensionality and noise. The fusion model achieves competitive performance by combining complementary

information from multiple modalities, although its performance is close to the multispectral model.

The analysis demonstrates that spectral data plays a crucial role in crop disease detection and that combining modalities can improve robustness. However, the effectiveness of fusion depends on the quality and preprocessing of hyperspectral data.

Future work may explore transformer-based multimodal fusion, larger datasets, and real-time deployment for precision agriculture.

6 References

References

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